

ALEX MERCER

Senior Software Debugging & Systems Diagnostics Engineer

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High-impact Software Diagnostics & Systems Debugging Specialist with 8+ years of experience analyzing complex distributed architectures, kernel-level crashes, memory leaks, and performance bottlenecks. Proven track record in root-cause analysis (RCA), static/dynamic code analysis, reverse engineering, and production incident remediation. Adept at creating specialized debugging toolchains and automated crash analysis pipelines to reduce Mean Time to Resolution (MTTR) by over 40%.

CORE COMPETENCIES & TOOLSET

Debuggers & Tracing: GDB, LLDB, WinDbg, Valgrind, Strace, Ltng, Perf, eBPF, Wireshark, DTrace

Languages & Runtimes: C, C++, Rust, Python, Go, x86-64 Assembly, Java (JVM Profiling)

Diagnostic Techniques: Core Dump Analysis, Heap & Stack Inspection, Race Condition Isolation, Memory Profiling, Static Analysis

Systems & Environments: Linux Kernel, POSIX Systems, Windows Internals, Microservices, Docker, Kubernetes

Automation & Scripting: GDB Python API, Bash, CI/CD Bug Triage Integrations, ELK Stack, Prometheus

PROFESSIONAL EXPERIENCE

Staff Systems Debugging Engineer | Apex Cloud Technologies 2023 – Present | San Jose, CA

- Production Incident Triage:** Serve as Tier-3 escalation lead for critical, non-reproducible bugs in a distributed Kubernetes microservices mesh serving 10M+ daily active users.
- Automated Crash Analysis:** Engineered an eBPF-based automated telemetry pipeline that captures micro-burst CPU spikes and memory corruption events, reducing MTTR for high-severity incidents by 45%.
- Memory Leak Resolution:** Identified and resolved a complex, non-deterministic memory leak in C++ RPC networking libraries via Valgrind and custom GDB scripts, saving \$120K annually in cloud infrastructure costs.
- Post-Mortem Leadership:** Authored detailed Root Cause Analysis (RCA) reports and mentored engineering teams on defensive programming practices and post-mortem debugging.

Senior Software Diagnostics & Performance Engineer | Nexus Systems Solutions 2020 – 2023 | Austin, TX

- Kernel & Driver Debugging:** Analyzed Linux kernel panic dumps and driver fault logs using Crash and GDB; fixed 30+ race conditions and deadlocks in low-level storage drivers.
- JVM & Application Profiling:** Utilized Async-Profiler and JProfiler to isolate high tail-latency issues in backend microservices, optimizing lock contention and improving API response times by 35%.
- Reverse Engineering:** Decompiled legacy proprietary binaries with Ghidra and IDA Pro to document undocumented API behaviors and patch critical security vulnerabilities without source access.

Software Engineer - Bug Remediation & Automation | Vanguard Software Corp 2018 – 2020 | Seattle, WA

- Static & Dynamic Analysis:** Integrated SonarQube and AddressSanitizer (ASan) into the automated CI/CD pipeline, catching 200+ memory safety issues before deployment to production.
- Root Cause Analysis:** Investigated multi-threaded concurrency bugs in core C++ processing engines using thread sanitizer (TSan) and custom logging harness.

KEY ACCOMPLISHMENTS & DEBUGGING PROJECTS

Custom GDB Extension Framework (Open Source)
Created a custom Python extension for GDB to automatically parse complex standard template library (STL) data structures in memory dumps, shortening manual investigation time for core dumps from hours to minutes.

EDUCATION & CERTIFICATIONS

B.S. in Computer Science

University of California, Berkeley (2014 – 2018)

Focus: Operating Systems, Systems Architecture

Certifications:

- Linux Foundation Certified Systems Administrator (LFCS)
- Certified Information Systems Security Professional (CISSP) - Auditing
- eLearnSecurity Certified Reverse Engineer (eCRE)